Soft Matter and Polymer Physics

Under constant revision- certain symbols might be poorly/incorrectly defined

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Basic Polymer models

Polymers are long chain-like, flexible structures of macromolecules. On large scales, their orientational correleations decay.

They can be “physically" described by tangent vectors along a path. In some coordinate frame where the position of a point along a path s is described by $\mathbf{r}$, the corresponding tangent vector $\mathbf{u}$ is:

$$\mathbf{u} = \frac{d\mathbf{r}(s)}{ds} \tag{1}$$

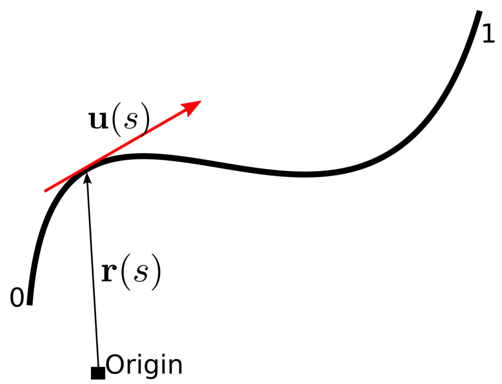

Figure 1: Schematic of a polymer chain in some reference frame. The path length varies between 0 and 1.

Since the orientational correleations are uncorreleated

$$\langle \mathbf{u}(s_1)\mathbf{u}(s_2)\rangle = \mathcal{C}(s_1, s_2) \tag{2}$$

where $\langle \ldots \rangle$ denotes the ensemble average over all the points and $\mathcal{C}$ is a constant function.

Aim: To formulate a Hamiltonian for these structures

Discrete Chain Model

Assumes a polymer to be made up of balls connected by some threads/springs (see Figure 2). Possible to describe the entire chain configuration by the position of the first bead $\mathbf{r}_1$ and the corresponding bond vectors for the subsequent beads $\mathbf{b}_i, \forall\, i \in \{1, \ldots, (N-1)\}$.

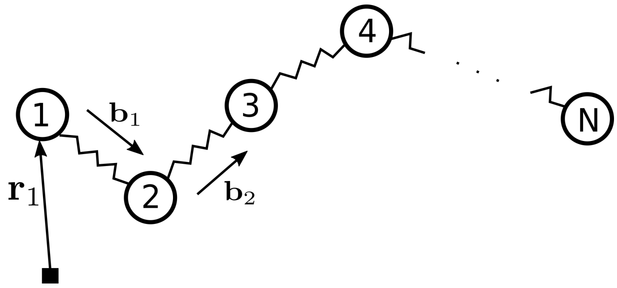

Figure 2: Schematic of a discrete chain polymer with N beads.